

1. Initial scalar-only prototype with $\cos(3)$ potential, introducing Z symmetry and kinks.

- **Suggested Lagrangian core (flat space, 3+1D):**

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \theta_4)^2 - \mu^2 \cos(3\theta_4) + \lambda(u^\mu u_\mu + 1) + \alpha P^{\mu\nu} \nabla_\mu u^\rho \nabla_\nu u_\rho + f(\theta_4) \nabla_\mu u^\mu + \mathcal{L}_\tau$$

2. Introduction of unit-timelike vector field u^μ with constraint term via .

Lagrangian (Minimal)

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \theta)^2 - \mu^2 \cos(3\theta) + \lambda(u^\mu u_\mu + 1) + \alpha \partial^\mu u^\nu \partial_\mu u_\nu + f(\theta) \partial_\mu u^\mu$$

3. Addition of kinetic term for $u^\wedge$: elastic-like dynamics, Einsteinian analogy.

② τ Fusion Algebra $\rightarrow$ Constraints on Correlators and Defect Dynamics

Fusion Rule Recap:

$$\tau_i + \tau_j + \tau_k \equiv 0 \pmod{3}$$

This implies:

- Triplet domain wall intersections must obey topological charge conservation.
- Fusion rules constrain allowable field configurations, especially at junctions or braiding events.

Constraints on Correlators:

Let $\mathcal{O}_{\tau_i}(x)$ be operators creating τ -defects. Then:

$$\langle \mathcal{O}_{\tau_1}(x_1) \mathcal{O}_{\tau_2}(x_2) \mathcal{O}_{\tau_3}(x_3) \rangle \propto \delta_{\tau_1 + \tau_2 + \tau_3 \equiv 0}$$

This is a **selection rule** emerging from the fusion algebra.

Encoding in **Lagrangian** or Path Integral:

1. Topological Action (Dijkgraaf–Witten-like):

$$S_\tau = \frac{2\pi i}{3} \int B \cup \delta A$$

where A is a discrete $\mathbb{Z}_3$ 1-form field, B is a 2-form Lagrange multiplier.

2. Effective Lattice Model:

Introduce τ field on links:

$$\mathcal{L}_{\text{fusion}} = \sum_{\Delta} \delta_{\tau_i + \tau_j + \tau_k \pmod{3}}$$

4. Coupling term $(\hat{u})u_-$ introduced: scalar-vector gradient interaction.

◆ MODEL 1: 1+1D θ_4 Domain Wall (Kink) Model

► Equation of Motion

From Lagrangian:

$$\mathcal{L}_\theta = \frac{1}{2}(\partial_\mu \theta)^2 - \mu^2 \cos(3\theta) \Rightarrow \theta''(x) = -3\mu^2 \sin(3\theta)$$

► Analytic Solution (Domain Wall Interpolating $0 \rightarrow 2\pi/3$):

$$\theta(x) = \frac{2}{3} \arctan \left(e^{\sqrt{3}\mu(x-x_0)} \right)$$

This solves the EOM under $\theta(-\infty) = 0$, $\theta(+\infty) = \frac{2\pi}{3}$.

► Energy Density Profile:

$$\mathcal{E}(x) = \frac{1}{2} \left(\frac{d\theta}{dx} \right)^2 + \mu^2 \cos(3\theta(x))$$

Peak localized at wall center x_0 , characteristic width $\sim 1/\mu$.

1 Optical Prediction via $\Delta n(\theta_4)$

Let $\theta_4(x)$ modulate refractive index:

$$n(x) = n_0 + \Delta n(\theta(x)), \quad \Delta n(\theta) = \eta \sin^2 \theta(x)$$

5. Implicit coupling via $()$: favoring fusion triplet formation at kinks.



Lagrangian Encoding via Discrete Topological Terms

Introduce $\mathbb{Z}_3$ 1-form gauge field A , valued in discrete group cohomology. Use **Dijkgraaf-Witten-like action**:

$$S_\tau = \frac{2\pi i}{3} \int_M B \cup \delta A$$

- A : 1-cochain field representing τ -sector boundary labels
- B : 2-cochain Lagrange multiplier enforcing **flatness constraint** $\delta A = 0$
- Forces local τ sums to vanish around each triangle (or junction)

Effect:

- Dynamically enforces $\tau_i + \tau_j + \tau_k \equiv 0$ at each interaction vertex
- Forbids forbidden τ -triplet configurations in path integral



Field Correlator Selection Rule

For τ -sector operators $\mathcal{O}_{\tau_i}(x)$:

$$\langle \mathcal{O}_{\tau_1}(x_1) \mathcal{O}_{\tau_2}(x_2) \mathcal{O}_{\tau_3}(x_3) \rangle \propto \delta_{\tau_1 + \tau_2 + \tau_3 \equiv 0}$$



3. UNIFIED LAGRANGIAN (Lattice-Suitable)

Let $A \in C^1(M, \mathbb{Z}_3)$ (τ field), $\theta \in C^0(M, \mathbb{R}/2\pi)$.



Action:

$$S = \int_M \left[\frac{1}{2} (\partial_\mu \theta)^2 - \mu^2 \cos(3\theta) + \lambda \cdot \delta_{\text{fusion violation}}(A) + \frac{2\pi i}{3} B \cup \delta A + \zeta A \cup d\theta \right]$$

Terms:

- Scalar dynamics and potential (θ_4 sector)
- τ fusion penalty
- Topological flatness constraint (BF/Dijkgraaf–Witten)
- Topological coupling of θ gradient to τ field

7. Full scalar-vector system with all terms active; precursor to constraint analysis.



SECTION I: MARK IV.2 LAGRANGIAN — CURRENT STRUCTURE



Core Fields

Field	Description	Domain	
$\theta_4(x)$	Real scalar field (angular misalignment)	$x \in \mathbb{R}^{1,3}$	
$u^\mu(x)$	Unit timelike vector field (wavefront flow)	$u^\mu u_\mu = -1$	
$\tau \in \mathbb{Z}_3$	Discrete twist variable	Defined over lattice edges (1-cochains)	



MARK IV.2 LAGRANGIAN (Continuum + Topological Hybrid Form)

$$\begin{aligned} \mathcal{L}_{\text{SAT}} = & \frac{1}{2}(\partial_\mu \theta_4)^2 - \mu^2 \cos(3\theta_4) \quad [\theta_4 \text{ dynamics}] \\ & + \lambda(u^\mu u_\mu + 1) + \alpha \nabla^\mu u^\nu \nabla_\mu u_\nu \quad [u^\mu \text{ constraint} + \text{kinetic}] \\ & + \beta \mathcal{C}^\mu(\theta_4) \nabla_\mu u^\nu u_\nu \quad [\theta_4\text{--}u^\mu \text{ coupling (under test)}] \\ & + \sum_{\triangle} \epsilon_f (1 - \delta_{\tau_i + \tau_j + \tau_k \bmod 3, 0}) \quad [\tau \text{ fusion penalty}] \\ & + \frac{2\pi i}{3} B \cup \delta A \quad [\text{topological } \mathbb{Z}_3 \text{ flatness}] \\ & + \zeta A \cup d\theta_4 \quad [\theta_4\text{--}\tau \text{ holonomy coupling (proposed)}] \end{aligned}$$

8. EulerLagrange derivations for θ_4 and u^μ showing dynamic coupling structure.

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📦 TERM STATUS

Term	Status	Notes
θ_4 kinetic + potential	Fixed	Standard sine-Gordon + $\mathbb{Z}_3$ vacua
u^μ constraint + dynamics	Stable	Projected kinetic term under ongoing formal variation
θ_4 – u^μ coupling (via $\mathcal{C}^\mu$)	Under Evaluation	Variational derivation in progress; constraints may generate effective curvature
τ fusion energy	Confirmed	Fusion constraint implemented via Monte Carlo; consistent with Dijkgraaf–Witten
$B \cup \delta A$	Stable	Encodes flatness of τ sector as topological gauge constraint
$A \cup d\theta_4$	Experimental	Coupling term linking domain walls to τ transitions; promising for defect binding

9. Constraint equations and extracted, defining primary/secondary constraints.

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$$\begin{aligned} \mathcal{L}_{\text{SAT}} = & \frac{1}{2}(\partial_\mu \theta_4)^2 - \mu^2 \cos(3\theta_4) \quad [\theta_4 \text{ dynamics}] \\ & + \lambda(u^\mu u_\mu + 1) + \alpha \nabla^\mu u^\nu \nabla_\mu u_\nu \quad [u^\nu \text{ constraint} + \text{kinetic}] \\ & + \beta \mathcal{C}^\mu(\theta_4) \nabla_\mu u^\nu u_\nu \quad [\theta_4\text{--}u^\nu \text{ coupling (under test)}] \\ & + \sum_{\triangle} \epsilon_f (1 - \delta_{\tau_i + \tau_j + \tau_k \bmod 3, 0}) \quad [\tau \text{ fusion penalty}] \\ & + \frac{2\pi i}{3} B \cup \delta A \quad [\text{topological } \mathbb{Z}_3 \text{ flatness}] \\ & + \zeta A \cup d\theta_4 \quad [\theta_4\text{--}\tau \text{ holonomy coupling (proposed)}] \end{aligned}$$

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10. Dirac bracket matrix built and rank-checked, confirming second-class system.



TASK D: STRESS-ENERGY TENSOR DERIVATION

From the **Lagrangian** (flat 1+1D):

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \theta_4)^2 - \mu^2 \cos(3\theta_4) + \lambda(u^\mu u_\mu + 1) + \alpha \nabla^\mu u^\nu \nabla_\mu u_\nu$$

Apply Noether procedure for energy-momentum tensor:

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\nu \phi - \eta^{\mu\nu} \mathcal{L}$$

(Sum over $\phi \in \{\theta_4, u^\mu\}$)

◆ For θ_4 sector (1+1D flat):

$$T_\theta^{\mu\nu} = \partial^\mu \theta_4 \partial^\nu \theta_4 - \eta^{\mu\nu} \left(\frac{1}{2}(\partial_\alpha \theta_4)^2 - \mu^2 \cos(3\theta_4) \right)$$

Evaluate on kink background:

$$\theta_4(x) = \frac{2\pi}{3} \cdot \frac{1 + \tanh(\mu x)}{2}$$

Then:

- T^{00} → energy density: localized bump centered on $x=0$
- T^{11} → pressure term: negative inside kink core (domain wall tension)

◆ For u^μ sector:

Approximate flat-spacetime contribution:

$$T_u^{\mu\nu} = \alpha \left(\nabla^\mu u^\lambda \nabla^\nu u_\lambda + \nabla^\lambda u^\mu \nabla_\lambda u^\nu - \frac{1}{2} \eta^{\mu\nu} \nabla^\alpha u^\beta \nabla_\alpha u_\beta \right)$$

Constraint term does **not** contribute to $T^{\mu\nu}$ (no derivatives).

11. Examination of stress-energy contributions from scalar and vector sectors.

Final $T^{\mu\nu}$ for u^μ Sector (Flat Background)

In 1+1D with $u^\mu = (u^0(t, x), u^1(t, x))$, the stress-energy tensor derived from the SAT Lagrangian:

$$T_u^{\mu\nu} = \alpha \left[(\nabla^\mu u^\lambda)(\nabla^\nu u_\lambda) + (\nabla^\lambda u^\mu)(\nabla_\lambda u^\nu) - \frac{1}{2} \eta^{\mu\nu} (\nabla^\alpha u^\beta \nabla_\alpha u_\beta) \right]$$

Output Components:

Component	Expression
T_u^{00}	$\alpha (2.5 \dot{u}_0^2 + 1.5 u_0'^2 + 1.5 \dot{u}_1^2 + 0.5 u_1'^2)$
$T_u^{01} = T_u^{10}$	$\alpha (\dot{u}_0 u_0' + \dot{u}_0 \dot{u}_1 + u_0' u_1' + \dot{u}_1 u_1')$
T_u^{11}	$\alpha (-0.5 \dot{u}_0^2 + 0.5 u_0'^2 + 0.5 \dot{u}_1^2 + 1.5 u_1'^2)$

Where:

- $\dot{u}_i = \partial_t u_i$
- $u_i' = \partial_x u_i$

 This form is ready for substitution into kink backgrounds or numerical evaluation.

Would you like the bracket matrix, constraint diagram, and $T^{\mu\nu}$ expressions exported to LaTeX or included in a Mathematica reference notebook? [\[>-\]](#)

12. Setup for Yukawa-like coupling: $(J^\mu)_-$ and discrete current logic.

Final $T^{\mu\nu}$ for u^μ Sector (Flat Background)

In 1+1D with $u^\mu = (u^0(t, x), u^1(t, x))$, the stress-energy tensor derived from the SAT Lagrangian:

$$T_u^{\mu\nu} = \alpha \left[(\nabla^\mu u^\lambda)(\nabla^\nu u_\lambda) + (\nabla^\lambda u^\mu)(\nabla_\lambda u^\nu) - \frac{1}{2} \eta^{\mu\nu} (\nabla^\alpha u^\beta \nabla_\alpha u_\beta) \right]$$

Output Components:

Component	Expression
T_u^{00}	$\alpha (2.5 \dot{u}_0^2 + 1.5 u_0'^2 + 1.5 \dot{u}_1^2 + 0.5 u_1'^2)$
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13. Full extended Lagrangian reflecting coupling across scalar, vector, and domains.

■ Derived Lagrangian (1+1D, Continuous Approximation)

We treat:

- θ_4 as a real scalar field
- τ as a lifted (approximate continuous) scalar field representing fusion violations over edges
- Coupling: fusion violation term weighted by θ_4 gradient

$$\mathcal{L} = \frac{1}{2}(\partial_t \theta)^2 - \frac{1}{2}(\partial_x \theta)^2 - \mu^2 \cos(3\theta) - \epsilon_f \tau^2 - \epsilon_f \gamma \tau^2 (\partial_x \theta)^2$$

📊 Euler–Lagrange Equations

1. For $\theta_4(\mathbf{x}, t)$:

$$\partial_t^2 \theta - \partial_x^2 \theta + 3\mu^2 \sin(3\theta) + 2\epsilon_f \gamma \tau^2 \partial_x^2 \theta + 4\epsilon_f \gamma \tau (\partial_x \tau)(\partial_x \theta) = 0$$

🧠 Interpretation:

- θ_4 propagates as a sine-Gordon-like field
- Coupled τ field modifies both effective stiffness and adds nonlinear backreaction
- Gradient "sees" τ fusion violations as topological friction or curvature sources

2. For $\tau(\mathbf{x}, t)$:

$$\frac{\delta \mathcal{L}}{\delta \tau} = -2\epsilon_f(1 + \gamma(\partial_x \theta)^2)\tau = 0$$

🧠 Interpretation:

- τ is energetically suppressed where θ_4 gradients are strong (e.g., near domain walls)
- τ condenses ($\tau \neq 0$) only where θ_4 is approximately uniform → consistent with τ crystallizing in domain interiors
- No kinetic term; τ is non-dynamical unless explicitly promoted